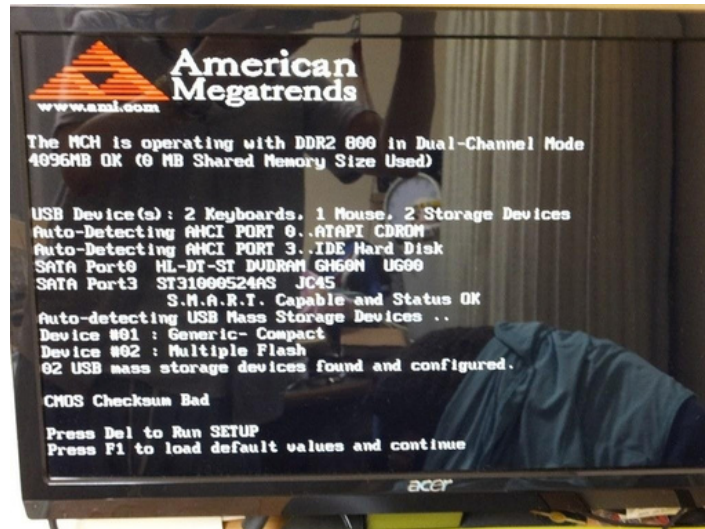


Task 1 - RAM Reseating Procedure



Real hardware photo: POST screen showing DDR2 memory detected

Real hardware photo: POST screen showing DDR2 memory detected

Opening a desktop or laptop should always be done with the system powered off and disconnected from the electrical supply. The purpose of reseating RAM is to remove and reinstall the memory module so that poor contact, dust, or an incompletely seated module can be ruled out as a cause of memory-related symptoms.

Step1 - Shutdownandisolatethesystem

Save any work, shut the computer down normally, disconnect the charger or AC power, and remove the battery if the model allows it. Press the power button for a few seconds after disconnecting power to discharge residual power. This reduces the chance of damaging components while handling the RAM.

Step2 - Openthecase

Remove the appropriate service panel using the correct screwdriver. Keep screws in a safe place. Before touching the internal components, discharge static electricity by touching a grounded metal part of the chassis or by using an antistatic wrist strap.

Step3 - LocatetheRAM

Identify the long DIMM slots on a desktop motherboard or the smaller SO-DIMM slot inside a laptop. The memory module is normally held by one or two retaining clips. Avoid touching the gold contacts on the module.

Step4 - Removethemodule

Push the retaining clips outward. The RAM should lift slightly from the slot. Hold the module by its edges and remove it without forcing it sideways. Check the slot and module for visible dust, dirt, damaged contacts, or broken retaining clips.

Step5 - ReinserttheRAM

Align the notch on the memory module with the key in the slot. The module will only fit in the correct orientation. Press down evenly on both ends until the retaining clips lock into position. Do not press on the center of the chip area.

Step6 - Reassembleandtest

Close the panel, reconnect power, and start the system. Check whether the computer completes POST, detects the expected memory capacity, and boots normally. If the system still shows memory errors, test one module at a time and, where available, try another compatible slot.

Observation and conclusion

A successful reseating operation should leave the module firmly locked and the system should recognize the installed memory. If the original symptom disappears after reseating, a loose connection or poor contact was a possible cause. If the symptom remains, further memory testing or component isolation is required.

Task 2 - GPU Troubleshooting Checklist

A faulty GPU can produce several different symptoms, so diagnosis should proceed from simple external checks toward component-level testing. The aim is to identify whether the problem is caused by the monitor, cable, power connection, GPU seating, software driver, overheating, or the graphics card itself.

1. Record the symptom

Note exactly what happens: no display, distorted graphics, colored lines, random crashes, black screen during gaming, driver errors, or unusually low graphics performance. Reproducing the symptom helps determine whether the fault is consistent.

2. Check the monitor and cable

Confirm that the monitor is powered on and set to the correct input. Reseat the HDMI, DisplayPort, or other display cable. If possible, test another cable, another monitor, or another output port. This separates a display-path problem from a GPU problem.

3. Inspect and reseat the GPU

Power the PC off and disconnect it from AC power. Check that the graphics card is fully seated in its PCIe slot and that its rear bracket is not preventing proper seating. Remove and reinstall the card only if necessary and safe to do so.

4. Check GPU power connections

Many dedicated graphics cards require one or more auxiliary PCIe power connectors. Confirm that the connectors are fully inserted and that the power supply is appropriate for the card. A loose power connector can cause black screens, crashes, or failure to initialize.

5. Check drivers and operating-system detection

After the physical checks, inspect the operating system's device information for the graphics adapter. A missing device, warning symbol, repeated driver failure, or driver reset can point toward a software or hardware problem. Reinstalling the correct driver can be used as a controlled test.

6. Check temperature and stability

Observe GPU temperature and fan behavior under normal use or a controlled workload. Excessive temperature, sudden fan failure, artifacts, or shutdowns under load can indicate cooling problems or a failing card. Avoid unnecessary stress testing when the system is unstable.

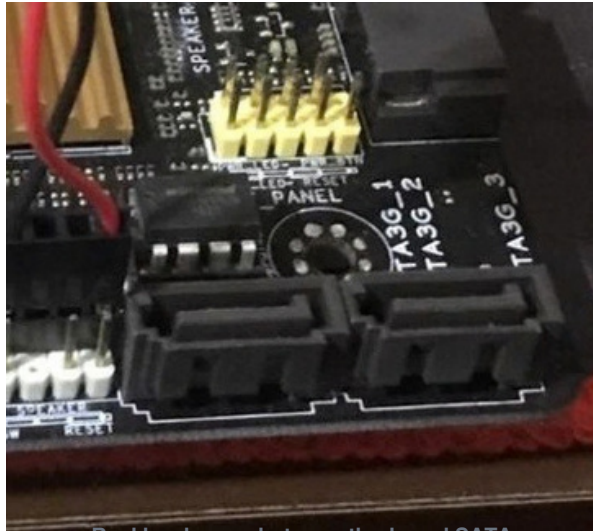
7. Cross-test to confirm

If available, test the system with integrated graphics or another known-good graphics card. If the system works normally with the alternate graphics solution, suspicion shifts toward the original GPU or its power path. If the same problem remains, investigate other components such as RAM, power supply, motherboard, or display equipment.

Final confirmation

The fix is confirmed only after the PC can boot, display correctly, run the required graphics workload, and remain stable for a reasonable period. A single successful boot is not enough to prove that an intermittent GPU fault has been solved.

Task 3 - Storage Issue Simulation



Real hardware photo: motherboard SATA connector area

Real hardware photo: motherboard SATA connector area

A storage connection problem can be simulated by disconnecting and reconnecting a SATA drive on a suitable test system. This should be performed only on a system that is powered off and safe to service. For an NVMe drive, the procedure is different because it uses an M.2 slot rather than a SATA data cable.

Before disconnecting

Record the existing condition first. Note whether the drive is visible in BIOS or UEFI, whether the operating system boots, and whether the drive appears in the system's storage information. This gives a baseline for comparison.

Disconnect the storage device

Shut the system down completely and disconnect AC power. For a SATA drive, disconnect the SATA data cable and, if the simulation requires it, the separate SATA power connector. Handle the connectors by their plugs rather than pulling on the cable.

Reconnect the drive

Reconnect the SATA data and power cables firmly. Inspect the connectors for damage and make sure the cable is not loose. If a second known-good SATA cable or motherboard port is available, it can be used for controlled comparison.

Check BIOS or UEFI

Start the system and enter BIOS or UEFI setup. Check whether the drive is listed among the detected storage devices. If the drive is missing, the issue may be related to power, cable connection, port selection, the drive itself, or motherboard storage configuration.

Observe operating-system behavior

If the drive is detected by firmware but the operating system reports a missing boot device, inaccessible volume, or similar message, the problem may involve boot configuration, partitions, filesystem condition, or the operating-system installation rather than the physical cable alone.

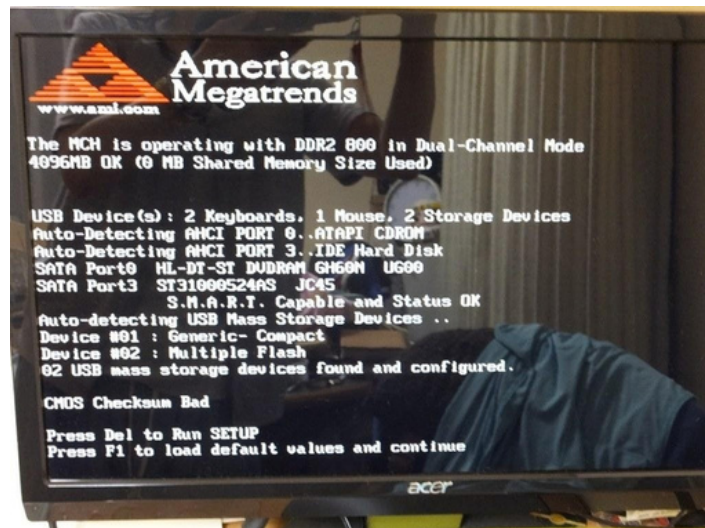
Expected before and after result

Before reconnection, a deliberately disconnected SATA drive may disappear from BIOS/UEFI or cause a boot-related error if it is the system drive. After reconnection, the drive should normally reappear in firmware and the system should return to its previous storage configuration if no other fault is present.

NVMe note

An NVMe SSD does not use a SATA data cable. It is installed directly into an M.2 slot and may be secured with a small screw. Therefore, the equivalent simulation would involve powering down, reseating the M.2 module, and checking detection in BIOS/UEFI.

Task 4 - Random Restart Diagnosis



Real hardware photo: CMOS Checksum Bad POST condition

Real hardware photo: CMOS Checksum Bad POST condition

For a PC that randomly restarts, the first priority is to identify whether the failure is related to memory, graphics, storage, power delivery, temperature, or the motherboard. A useful diagnostic process changes one variable at a time so that the result can be linked to a specific component or condition.

Initial symptom check

Record when the restart occurs. If it happens during gaming or other heavy graphics work, GPU temperature and power delivery become important checks. If it happens during ordinary desktop use, memory, power supply, motherboard, or software-related causes should also be considered.

RAMcheck

Power the PC off and reseal the RAM. If more than one memory module is installed, test one module at a time using the recommended motherboard slot. Check whether the restart pattern changes. Memory-related instability can sometimes appear as freezes, blue screens, failed POST, or sudden restarts.

GPU check

Inspect the graphics card seating, auxiliary power connectors, display behavior, and cooling. If the restart occurs specifically under graphics load, compare behavior with integrated graphics or another known-good GPU when available. A GPU fault is more strongly suspected when the issue disappears during a controlled graphics cross-test.

Storage check

Check whether the storage device is consistently detected in BIOS/UEFI and whether the operating system reports disk errors. Storage faults more commonly cause boot failures, application errors, corrupted files, or system freezes than a clean sudden restart, so storage should be evaluated alongside the power and thermal path.

Power and temperature check

Random restarts under load can also be associated with an inadequate or unstable power supply or excessive temperature. Inspect power connectors and cooling, and monitor temperatures using an appropriate diagnostic method. Do not continue stressing hardware that is overheating or showing signs of electrical instability.

Controlled isolation

Change only one component or condition at a time. For example, test one RAM module, then another; use integrated graphics instead of the dedicated GPU; or test with a known-good power supply if available. This avoids making several changes at once and losing the ability to identify the cause.

Finalconfirmation

The suspected component should only be considered the cause after a repeatable test shows that the symptom follows that component or disappears when it is replaced or bypassed. Record the test condition and result so the troubleshooting process can be explained clearly.

Practical Task Reference

The following image is the task sheet supplied for this assignment. It is included only as a practical reference for the questions answered above.

Source: task sheet supplied with the assignment.